(5) The structure of polytopal complexes is far from understood. Even polytopal subdivisions of 3-polytopes pose more problems than there are answers in the moment. The reader will find some open problems among the exercises; we refer also to Bayer [60, 61] and to Lee [354, 357]. The space of all regular subdivisions of a polytope will reappear in the next lecture, in a completely different setting. There are challenging questions in the study of f-vectors of subdivisions and their properties as well. We just mention Stanley's *local h-vector* as a new tool; see Stanley [520] and Chan [145].

Problems and Exercises

8.0 Show that every polyhedral subdivision of a 2-polytope is extendably shellable: We can start with an arbitrary 2-face, and never get stuck. (This is classical, see Newman [422], but also [173] and [239].) Show that one cannot, however, prescribe the last 2-face of a shelling.

8.1 (i) Show that a set of facets of the d-cube determines a shellable subcomplex of ∂C_d if and only if it contains no facets (is empty), or all facets, or if it contains at least one facet such that the opposite facet is not in the complex. Deduce that the boundary complexes of the d-cubes are extendably shellable.

(ii) Describe a shelling of the d-dimensional crosspolytope. Use it to compute the f-vector and the h-vector of the d-dimensional crosspolytope.

(iii) Given the h-vector of a simplicial polytope P, how can one derive the h-vector of the bipyramid $\mathrm{bipyr}(P)$?

(iv) Verify that the d-dimensional crosspolytopes ${C_d}^{\Delta}$ are extendably shellable for $d \leq 4$. (Surprisingly, they are *not* extendably shellable for $d \geq 12$, as proved by Hall [269]!)

8.2 Show that an ordering $F_1, F_2, \ldots, F_s$ of the facets of a pure simplicial complex is a shelling order if and only if for every $i \geq 1$, the facet F_i contains a unique minimal face which is not contained in an earlier facet F_j with $j < i$.

Show that the permutation $F_1, \ldots, F_t$ is a shelling of Δ if and only if Δ is partitionable with a partition such that

$$R_i \subseteq F_j \qquad \Longrightarrow \qquad i \leq j.$$

8.3* Is every d-diagram shellable? What can you say about the case $d = 3$? (For $d = 2$ this is true, by Exercise 8.0. If you want a guess for $d \geq 3$, I'd vote for "no," because of the rule of thumb, "if Bruggesser-Mani doesn't shell it, then it isn't shellable.")

8.4* If a polytopal complex $\mathcal{C}$ is shellable, but not necessarily simplicial, is it still true that its stars and links are shellable?
(Be careful: results of Pachner [433] show that the boundary of a shellable ball need not be shellable. The answer is "yes" for stars according to Courdurier [163], but it might still be "no" for the links.)

8.5 Show that for every vertex v of a d-polytope P, there is a polytopal complex $\mathcal{C}$ that subdivides the boundary complex of the vertex figure, $|\mathcal{C}| = P/v$, and that is combinatorially equivalent to $\text{link}(v, \mathcal{C})$.
(The face fan of the complex $\mathcal{C}$ is a *flattening* of the boundary complex of P at the vertex v; see MacPherson [373].)

8.6 If $\mathcal{C} = \mathcal{C}(\partial P)$ is the boundary complex of a (nonsimplicial) polytope and v is a vertex of P, then $\text{link}(v, \mathcal{C})$ is shellable. To prove this, show that the links are isomorphic to the boundary complexes of polytopes. (Hint: Use a point beyond v.)

8.7* What is the smallest possible number of vertices for a nonshellable triangulation of a 3-polytope?
(Rudin [468] claims that one needs 14 vertices if the complex is realizable as a simplicial geometric subdivision of a simplex. Is that true?)

How many vertices are needed for a simplicial, nonshellable 3-sphere?

8.8* Beat Lemma 8.6: is this the smallest pile of cubes that is not extendably shellable?

8.9 Every shelling $F_1, F_2, \ldots, F_s$ of the facets of a polytope P also induces, for every facet F_i, an ordering of the facets of F_i. Namely, one can take facets of F_i in the order in which they appear in the list $F_1 \cap F_i, \ldots, F_{i-1} \cap F_i, F_{i+1} \cap F_i, \ldots, F_s \cap F_i$. A shelling is *perfect* if this ordering is a shelling order of the boundary of F_i, for all i.
For example, shellings of simplicial polytopes are always perfect.

(i) Show that Bruggesser-Mani shellings are not perfect in general.
(ii) Show that the d-cubes C_d have perfect shellings, for all $d \geq 1$.
(iii) Show that all 3-polytopes have perfect shellings.
(iv) Show that the polars of cyclic polytopes $C_d(n)^\Delta$ have perfect shellings.
(v)* Does every polytope have a perfect shelling? (Kalai)

8.10 For a d-polytope P, show that the linear functions on P^Δ in general position really correspond to the Bruggesser-Mani line shellings of P.

If P is simplicial, verify that under this correspondence the vertices v_j of in-degree k on P^Δ correspond to the facets with restriction set of size $|R_j| = d - k$.

In particular, the formula

$$f^O \;=\; h_0^O + 2h_1^O + 4h_2^O + \ldots + 2^k h_k^O + \ldots + 2^d h_d^O,$$

which we used for the total number of faces of P^Δ in Section 3.4, amounts to the evaluation of $\boldsymbol{f}(1) = \boldsymbol{h}(2)$ for the polytope P.

Using this, show that Kalai's "good orientations" on the graph $G(P^\Delta)$ are exactly the shelling orders for ∂P.

8.11 Prove the upper bound theorem for simple d-polytopes with n facets. For this, consider a linear function $\boldsymbol{c} \in (\mathbb{R}^d)$ in general position on a simple d-polytope $P \subseteq \mathbb{R}^d$ with n facets. For $t \in \mathbb{R}$ define $h_k{}^\Delta(P,t)$ to be the number of vertices in $\{\boldsymbol{x} \in P : \boldsymbol{cx} \leq t\}$ that are the highest point for k different edges (as suggested by the previous exercise).

By letting t increase, show that for all $t \in \mathbb{R}$,

$$(d{-}k)h_k{}^\Delta(P,t) + (k{+}1)h_{k+1}{}^\Delta(t) \;=\; \sum_F h_k{}^\Delta(F,t) \;\leq\; n\, h_{k+1}{}^\Delta(P,t),$$

where the sum is over all the n facets of P, and the second inequality follows from consideration of a linear function $\boldsymbol{c}$ for which the vertices of F are smaller than all other vertices of P. Deduce from this that

$$h_k{}^\Delta(P) \;\;\leq\;\; \binom{n-k-1}{d-k},$$

for $d \geq k \geq d - \lfloor \frac{d}{2} \rfloor$, and from this the upper bound theorem.

(This is the "dual proof" of the upper bound theorem 8.23, from McMullen [389, Note added in proof].)

8.12 For a simple d-polytope $P \subseteq \mathbb{R}^d$ with n vertices, a numbering of the vertices by $1, 2, \ldots, n$ is called *completely unimodal* if every k-face $(2 \leq k \leq d)$ has a unique local minimum, that is, every face F has only one vertex such that all its neighbors on F get a larger number.

(i) Show that the completely unimodal numberings of P exactly correspond to the shelling orders of the (simplicial) polar polytope P^Δ (Williamson Hoke [566, Prop. 1]; see also [279]).

(ii) Show that if P is the d-dimensional cube, then it suffices to assume that the numbering has a unique local minimum on every 2-face. (Hammer, Simeone, Liebling & de Werra [270]; see [566, Prop. 2]).

(iii) Show that the 4-cube has a numbering that is not completely unimodal, but for which every k-face with $k \neq 2$ has a unique local minimum.

(iv) Construct a simple 3-polytope P with a numbering such that every 2-face has a unique local minimum, but there are two local minima on P.
(Hint: First construct an acyclic orientation of a 3-polytopal graph such that there are two sinks, but only one sink if one restricts to one of the facets.)

8.13 Let $f = (1, 23, 47, 52, 38, 12)$.

(i) Is f the f-vector of a simplicial complex?

(ii) Is f the f-vector of a shellable complex?

(iii) Is f the f-vector of a simplicial polytope?

8.14 Show that $\partial_k(n)$ is a monotonically increasing function of n (for fixed k).
Characterize the values n for which $\partial_k(n) = \partial_k(n+1)$.

8.15 Characterize the f-vectors of connected simplicial complexes, as follows. The sequence $\boldsymbol{f} = (1, f_0, f_1, f_2, \ldots, f_d)$ is the f-vector of a connected simplicial complex if and only if it satisfies the conditions of the Kruskal-Katona Theorem 8.32 and the additional relation $\partial^3(f_2) \leq f_1 - f_0 + 1$.
(This is due to Björner [91].)

8.16* Characterize the f-vectors of pure simplicial complexes.

(i) Start in low dimensions, with 2-dimensional and 3-dimensional simplicial complexes. Note that there are gaps: for example, a 2-dimensional pure simplicial complex with $f_2 = 4$ facets has at least $f_1 \geq 6$ edges, and at most $f_1 \leq 12$ edges — but $f_1 = 7$ is impossible.
(See Leck [351].)

(ii) In general, this is probably an intractable problem, since a complete answer would solve virtually all basic problems in design theory — this observation may originally be due to Singhi & Shrikhande [501, p. 67]: it is called “trivial” there. (See [67] for more on design theory.)
As an example, show that the existence of a projective plane of order d is equivalent to the existence of a pure simplicial complex (of dimension d) with $f_0 = d^2 + d + 1 = f_d$ and $f_1 = \binom{f_0}{2}$.
(It is a notorious problem to show that such an object can exist only if d is a power of a prime. The nonexistence is classical for $d = 6$. It was only recently proved for $d = 10$, by Lam, Thiel & Swiercz and a CRAY 1A [348]. It is completely open for $d = 12$. Don't try! Try!)

8.17 For the standard octahedron $C_3{}^\Delta \subseteq \mathbb{R}^3$, find a shelling line $\ell \in \mathbb{R}^3$ that generates the shellings of Example 8.22.
(Hint: Use Exercise 8.10).

Show that the facet ordering of the octahedron (labeled as in Example 8.22)

$$123, \quad 234, \quad 345, \quad 135, \quad 246, \quad 126, \quad 156, \quad 456,$$

is a shelling order for the octahedron. Prove that, however, it is not a Bruggesser-Mani shelling for *any* realization of the octahedron.
(Smilansky [504])

8.18 Derive

$$f_{k-1} \quad = \quad \sum_{i=k}^{d} (-1)^{d-i} \binom{i}{k} f_{i-1} \qquad \text{for } k = 0, 1, \ldots, d$$

for the f-vector of a simplicial polytopes from the Dehn-Sommerville equations of Theorem 8.21.

In particular, how does the "obvious" equation $2f_{d-2} = df_{d-1}$ follow from the Dehn-Sommerville equations?

8.19 Prove Stanley's trick: give a formula for the (i, j)-entry of Stanley's difference table, and show that the last row correctly computes the h-vector.
Why are all the entries of the table nonnegative for a shellable complex?

8.20 (i) Prove that for $0 \leq j \leq k \leq d$, one has

$$\sum_{i=j}^{d} \binom{d-i}{k-i} \quad = \quad \binom{d+1-j}{d+1-k}.$$

(ii) Prove directly that for $0 \leq j \leq d < n$, we have

$$\sum_{i=0}^{j} (-1)^{j-i} \binom{d-i}{d-j} \binom{n}{i} = \binom{n-d-1+j}{j}.$$

For this, verify that the equation is true for $d = j$ (by induction) and for $j = 0$, and then use induction on d, where the left-hand side and the right-hand side satisfy the same simple recursion. (See [252, p. 149], and also [240, p. 169].)

Use this to compute the h-vector for simplicial neighborly polytopes, directly from Definition 8.18.

8.21 Use Exercise 2.20 to show that one need not consider unbounded polyhedra for the upper bound theorem, as follows.

For every unbounded d-polyhedron P with at least two vertices, there exists a d-polytope with the same number of facets, but with more vertices than P.

What happens in the cases where P has at most one vertex?

8.22 Give natural bijections between

- the k-multisets with elements from $[n]$,
- the monomials of degree k in the variables $x_1, x_2, \ldots, x_n$, and
- vectors $\boldsymbol{z} \in \mathbb{N}_0^n$ with $\mathbb{1}\boldsymbol{z} = k$.

Show that under these bijections, inclusion of multisets corresponds to divisibility of monomials and to the componentwise ordering of vectors.

Give three more proofs of $\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}$. For example,

(i) Show that every k-multiset with elements in $[n]$ corresponds to a sequence like $**|*||***|*|$ with n stars and $k{-}1$ bars, where the number of stars between the ith bar and the $(i{-}1)$st bar is the multiplicity of i in the multiset. Then count the star-bar strings.

(ii) Use induction on n, and a basic identity for binomial coefficients.

For inspiration, see also Stanley [517, Sect. 1.2].

8.23 On the vertex set $[3n] = \{1, 2, \ldots, 3n\}$, consider the pure complex of dimension $d = 3n-4$ generated by the n facets $[3n]\backslash\{3i{-}2, 3i{-}1, 3i\}$ for $i = 1, \ldots, n$. There are 3^n minimal nonfaces: the sets of cardinality n that contain exactly one of $3i{-}2, 3i{-}1, 3i$ for each i.

By comparing this complex to an $(n{+}1)$-neighborly one, show that we have

$$
\begin{array}{ll}
f_{i-1} = \binom{3n}{i} & h_i = \binom{i+3}{3} \ \text{ for } i < n \\
f_{n-1} = \binom{3n}{i} - 3^n & h_n = \binom{n+3}{3} - 3^n \\
f_n \ = \binom{3n}{i} - n{\cdot}3^n & h_{n+1} = \binom{n+4}{3} + (n-4)3^n
\end{array}
$$

so that for $n \geq 6$ we have $h_n < 0$, and the upper bound condition of Lemma 8.26 is violated for h_{n+1} (where $n+1 \leq \lfloor \frac{d}{2} \rfloor$) for a pure complex. (Wistuba & Ziegler [567])

8.24 Show that the k-skeleta of d-polytopes are shellable polyhedral complexes.

(i) Show that r-lex order defines a shelling order $F_1(k), F_2(k), \ldots$ for the $(k-1)$-skeleton of the d-simplex, by directly verifying condition 8.1(ii′).

(ii) Show that in fact the k-skeleta of all shellable polyhedral complexes are shellable.

(iii)* Is the $(k-1)$-skeleton of every d-simplex extendably shellable? (This is the *shelling extension conjecture*, due to Simon [500, Ch. 5]; the conjecture is known to be true for $k \leq 3$, by Björner & Eriksson [92], and for $k \geq d-1$, by Kalai.)

8.25 Show the following weaker version of Macaulay's theorem, which estimates the M-sequence $\boldsymbol{h} = (h_0, h_1, \ldots, h_d)$ without using the subtle operator ∂^k. If $h_k = \binom{x}{k}$ for some real $x \in \mathbb{R}$ and if $k \geq 1$, then $h_{k-1} \geq \binom{x-1}{k-1}$.
(Björner, Frankl & Stanley [93, Thm. 3])

8.26* What combinatorial conditions on a simplicial complex imply that $h_i \leq \binom{n-d-1+i}{i}$?
(The result is known for shellable complexes, but only with algebraic tools, like Kalai's "algebraic shifting." Is there a fully combinatorial rule that with every shellable complex would associate a multicomplex whose f-vector is the h-vector of the complex? Can one prove it for pure complexes satisfying the Dehn-Sommerville equations, like Eulerian complexes [324] [62] ?)

8.27 Show that the maximal number of vertices of a d-polytope with $2d$ facets is larger than the number of vertices of the d-cube, for $d \geq 4$. For large d, the maximal number is roughly

$$\binom{3\lfloor\frac{d}{2}\rfloor}{\lfloor\frac{d}{2}\rfloor} \approx \left(\frac{27}{4}\right)^{\lfloor\frac{d}{2}\rfloor},$$

considerably more than 2^d.

8.28 Show that the f-vectors of general 3-polytopes are exactly the vectors of the form

$$\begin{aligned} \boldsymbol{f}(P_3) &= (1,4,6,4) + a(0,1,1,0) + b(0,0,1,1), \\ &\qquad \text{with } 2a \geq b \geq 0,\ 2b \geq a \geq 0, \end{aligned}$$

and the f-vectors of simplicial 3-polytopes are given by $b = 2a$, i.e.,

$$\boldsymbol{f}(P_3) = (1,4,6,4) + g_1(0,1,3,2), \quad \text{with } g_1 \geq 0.$$

8.29* Characterize the f-vectors of d-polytopes.

(The f-vectors for 3-polytopes were characterized by Steinitz [526]: see the previous exercise. This is a big unsolved research problem for every $d \geq 4$. See Ehrenborg [193] for a recent account.
For $d = 4$, much more is known. For example, the possible pairs (f_i, f_j) have been characterized for all $i < j$ (see Bayer [59], Bayer & Lee [63, Sect. 3.8], and Höppner & Ziegler [278]). According to [578] the *fatness* parameter $F(P) := \frac{f_1+f_2-20}{f_0+f_1-10}$ plays a crucial role: Can it be arbitrarily large? Polytopes with $F(P)$ arbitrarily close to 9 were constructed in [579], with further analysis in [472].)

8.30 Prove Björner's Theorem 8.39.
For the first half, you need a lemma that verifies the monotonicity of the rows of M_d, and shows that the peak lies between $j = \lfloor \frac{d}{2} \rfloor$ and $j = \lfloor \frac{3(d-1)}{4} \rfloor$. For the second half, you can use g-vectors of the form

$$\boldsymbol{g} \quad = \quad \left(1, g_1, \left(\!\!\binom{g_1}{2}\!\!\right), \left(\!\!\binom{g_1}{3}\!\!\right), \ldots, \left(\!\!\binom{g_1}{k}\!\!\right), 0, \ldots, 0\right),$$

such that the kth row of M_d peaks at m_{kp}. Now let g_1 get very large.

8.31 Let P be a simplicial d-polytope P, and let P' be obtained by a stellar subdivision (as defined in Exercise 3.0: erecting a "pyramidal cap" over a facet). Show that

$$\boldsymbol{f}(\partial P') \quad = \quad \boldsymbol{f}(\partial P) \ + \ \boldsymbol{f}(\partial \Delta_d) \ + \ \boldsymbol{f}(\partial \Delta_{d-1}) \ - \ 2\boldsymbol{f}(\Delta_{d-1}).$$

8.32 Consider the polytopes $C_d(n)^{<r>}$ obtained by r stellar subdivisions from cyclic ones. Using the previous exercise, show that the g-vector of $C_d(n)^{<r>}$ is given by $(1, g_1+r, g_2, \ldots, g_{\lfloor \frac{d}{2} \rfloor})$, where $g_i = \binom{n+d-2+i}{i}$ represents the g-vector of the cyclic polytope $C_d(n)$.

8.33* Are the f-vectors of (general) polytopes unimodal for $d \leq 7$?
(Connected sums of the form $P \# P^{\Delta}$ have unimodal f-vectors in dimension $d \leq 7$, see Björner [90, Sect. 3].)

Similarly, what is the smallest number of vertices for a d-polytope with nonunimodal f-vector?

(Eckhoff [188] has nonunimodal f-vectors for 8-polytopes with 6375 vertices and for 9-polytopes with only 1393 vertices (Example 8.41). Can you do with less? In the case of simplicial polytopes, Eckhoff proved $d_0 = 19$, but the smallest number of vertices is not known, either: here $f_0 = 1320$ is the current record, see Example 8.40.)

8.34 For simplicial d-polytopes, show that $f_k < f_{d-2-k}$ and $f_k \leq f_{d-1-k}$, for $0 \leq k \leq \lfloor \frac{(d-3)}{2} \rfloor$. (Björner [86, 90])

8.35* Does $f_0 < f_1 < f_2 < \ldots < f_{\lfloor d/4 \rfloor}$ hold for all d-polytopes?

8.36* Every centrally symmetric d-polytope has at least 3^d proper faces.

(This is known in the case of simplicial and of simple polytopes, proved by Stanley [518]. However, even to show that every simplicial centrally symmetric polytope has at least 2^d facets — a fact first proved by Bárány & Lovász [38] — one knows of no simple, "elementary" argument. Centrally symmetric d-polytopes with exactly 3^d proper faces exist: take for example the d-cubes, and all the polytopes which one can construct from k-cubes by taking products and polars. Kalai [302] conjectures that this yields all the polytopes with exactly 3^d proper faces.)

8.37* For every integer $k \geq 1$, is there an integer $f(k)$ such that every d-polytope with $d \geq f(k)$ has a k-face that is either a simplex, or combinatorially equivalent to a k-cube?

For every integer $k \geq 1$, is there an integer $g(k)$ such that every d-polytope with $d \geq g(k)$ has a *quotient* that is a k-simplex, that is, it has faces $G_1 \subseteq G_2$ such that $[G_1, G_2] \cong L(\Delta_k) = B_k$ (that is, the k-simplex arises as an iterated vertex figure of a face; cf. Exercise 2.9)?

(The first question is due to Gil Kalai [303], who even conjectures that "in some sense" a "typical" k-face of a "typical" simple d-polytopes with n facets will be combinatorially equivalent to the k-cube, if d and $n-d$ are both large enough compared with k.
In [303], Kalai proves that $f(2)$ is finite — in fact, $f(2) = 5$.
The second question is due to Micha Perles [438], who remarks that $g(0) = 0$, $g(1) = 1$, $g(2) = 3$ (by Euler's equation), and $g(3) \geq 5$ (from the 24-cell).)

8.38 Investigate the face numbers of cubical d-polytopes.
In particular, show the following:

(i) Every 3-dimensional cubical polytope has more vertices than facets (in fact, $f_2 = f_0 - 2$).

(ii) If P is a cubical zonotope with n zones, then

$$f_0 = 2\Big(\tbinom{n-1}{0} + \tbinom{n-1}{1} + \ldots + \tbinom{n-1}{d-1}\Big), \quad f_{d-1} = 2\tbinom{n}{d-1}.$$

In particular, P has more vertices than facets for $d \geq 3$.

(iii) If P is a cubical d-polytope, then $f_k(P) \geq f_k(C_d)$ holds for all k. (Blind & Blind [102])

(iv) Study the possible f-vectors of 4-dimensional cubical polytopes. In particular, they can have more facets than vertices. (Jockusch [291])

(v)* Does every cubical d-polytope with $d \geq 4$ have an even number of vertices? (For even d this was shown by Blind & Blind [104].)

8.39* Is every cubical polytope rational?

8.40 The *twisted lexicographic ordering* on the facets of $C_d(n)$ is defined as follows. Consider facets $F = \{i_1, \ldots, i_d\}$ and $G = \{j_1, \ldots, j_d\}$, and let k be the smallest index where they differ, $i_k \neq j_k$. Setting $i_0 = j_0 = 0$, define that $F \prec' G$ holds either if $i_k < j_k$ and $i_{k-1} = j_{k-1}$ is even, or if $i_k > j_k$ and $i_{k-1} = j_{k-1}$ is odd. Otherwise we set $F \succeq' G$.

(i) Show that every facet is adjacent to the previous one.
(ii) Show that $\prec'$ is a shelling order for $C_d(n)$, if $d \leq 4$.
(iii) Show that $\prec'$ is not a shelling order in general.
(Hint: list the first 8 facets in the ordering for $C_7(10)$.)

(This linear ordering is from Gärtner, Henk & Ziegler [219, Sect. 5], motivated by Klee's construction [325, Thm. 1.1] of a Hamilton cycle in the graph of $C_d(n)^\Delta$. Part (iii) was observed by Robert Hebble.)

8.41* Show that for $n \geq 8$ the cyclic polytope $C_4(n)$ cannot be realized in such a way that it has a Bruggesser-Mani shelling for which every facet is adjacent to the previous one. Equivalently, the polar $C_4(n)^\Delta$ cannot be realized with a monotone path through all the vertices. (Indeed, Pfeifle [441] verified that for $8 \leq n \leq 12$ there is no Hamilton path that would satisfy the known *combinatorial* conditions:

(i) induce a unique sink on each face [566] and
(ii) satisfy the Holt-Klee condition [282] that there are d vertex-disjoint graphs from source to sink in any orientation of the graph of a simple d-polytope that is induced by a linear function.

Thus it is an entirely combinatorial problem to show that no such ordering on the vertices of $C_4(n)^\Delta$ exists for any $n \geq 8$. There is no dual-to-neighborly polytope of dimension $d = 6$ with $n = 9$ facets and an monotone path through all its $f_0(C_6(9)^\Delta) = 30$ vertices [443]. Thus, in general $M(d, n)$ is smaller than the value $f_0(C_d(n)^\Delta)$ provided by the upper bound theorem. Compare Problem 3.11*.)

8.42* It seems to be an *open problem* to show that all f-vectors of cyclic polytopes are unimodal. Are they?

8.43 For $n > d \geq 2$, a *stacked polytope* $\mathrm{St}_d(n)$ is a simplicial d-polytope with n vertices that is obtained by starting with a d-simplex and successively adding $n - d - 1$ vertices "beyond a facet."

(i) Show that every stacked polytope $\mathrm{St}_d(n)$ is a connected sum of d-simplices.
(ii) Show that every polytope that is combinatorially isomorphic to $\mathrm{St}_d(n)$ is a stacked polytope. (In the terminology of [459], this is since the glueing simplex is "necessarily flat.")
(iii) Compute the f-vector of $\mathrm{St}_d(n)$, and show that for each fixed $d \geq 2$, it grows linearly with n.
(iv) Show that for every $d \geq 3$, the number of combinatorial types stacked polytopes $\mathrm{St}_d(n)$ grows exponentially with n.